

Trigger level ranges

Any input channel	±8 divisions from center of screen, ±8 divisions from 0 V when vertical LF reject trigger coupling is selected
Aux In (External)	±8 V
Line	The line trigger level is fixed at about 50% of the line voltage.

Trigger frequency readout

Provides 6-digit frequency readout of triggerable events.

Trigger types

Edge	Positive, negative, or either slope on any channel. Coupling includes DC, AC, HF reject, LF reject, and noise reject.
Sequence (B-trigger)	Trigger Delay by Time: 9.2 ns to 8 s. Or Trigger Delay by Events: 1 to 4,000,000 events. Not available when "Either" edge is selected.
Pulse Width	Trigger on width of positive or negative pulses that are >, <, =, ≠, or inside/outside a specified period of time.
Timeout	Trigger on an event which remains high, low, or either, for a specified time period (4 ns to 8 s).
Runt	Trigger on a pulse that crosses one threshold but fails to cross a second threshold before crossing the first again.
Logic	Trigger when any logical pattern of channels goes false or stays true for specified period of time. Any input can be used as a clock to look for the pattern on a clock edge. Pattern (AND, OR, NAND, NOR) specified for all input channels defined as High, Low, or Don't Care.
Setup and Hold	Trigger on violations of both setup time and hold time between clock and data present on any of the analog and digital input channels.

Setup and hold trigger type	Description
Setup Time Range	-0.5 ns to 1.024 ms
Hold Time Range	1.0 ns to 1.024 ms
Setup + Hold Time Range	0.5 ns to 2.048 ms

Rise/Fall Time	Trigger on pulse edge rates that are faster or slower than specified. Slope may be positive, negative, or either and time range is 4.0 ns to 8 s.
Video	Trigger on all lines, odd, even, or all fields on NTSC, PAL, and SECAM video signals. 480p/60, 576p/50, 720p/30, 720p/50, 720p/60, 875i/60, 1080i/50, 1080i/60, 1080p/24, 1080p/24sF, 1080p/25, 1080p/30, 1080p/50, 1080p/60 Custom bi-level and tri-level sync video standards.
Parallel (available when option 3-MSO is installed)	Trigger on a parallel bus data value. Parallel bus can be from 1 to 20 bits (from the digital and analog channels) in size. Binary and Hex radices are supported.

Acquisition system**Acquisition modes**

Sample	Acquire sampled values.
Peak Detect	Captures glitches as narrow as 1.5 ns (1 GHz models), 2.0 ns (500 MHz models), 3.0 ns (350 MHz models), 5.0 ns (200 MHz models), 7.0 ns (100 MHz models) at all sweep speeds
Averaging	From 2 to 512 waveforms included in average.
Envelope	Min-max envelope reflecting Peak Detect data over multiple acquisitions. Number of waveforms in the envelope selectable between 1 and 2000 and infinity
Hi Res	Real-time boxcar averaging reduces random noise and increases vertical resolution.
Roll	Scrolls waveforms right to left across the screen at sweep speeds slower than or equal to 40 ms/div.